DIAGONAL: Entity-Grounded Topological Evaluation for Multi-Shot Video Narrative Compliance

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Abstract

Text-to-multi-shot video (T2MSV) generation models are evaluated primarily on visual quality and frame-level consistency, yet these metrics fail to capture whether models can execute *narrative dynamics*—the structured entry and exit of entities across shots. We formalize multi-shot narratives as *state transition matrices* $\Delta S \in \{-1, 0, +1\}^{(T-1)\times N}$, where each element encodes an atomic operation: entry (+1), exit (−1), or hold (0). We prove that all possible 3-shot narrative topologies can be spanned by just two basis matrices under column permutation and linear combination (**Theorem 1**), establishing mathematical completeness. Building on this, we introduce DIAGONAL, a benchmark comprising 1,000 prompts spanning 7 narrative patterns (3 basis, 4 derived). A vision-language model constructs the observed presence matrix S_{obs} via per-entity binary queries, yielding a clean matrix-matching pipeline: the *Prescribed Transition Match* (PTM) measures the fraction of structurally prescribed transitions ($\Delta S \neq 0$ cells) correctly observed in the generated video. Evaluation of four SOTA T2MSV architectures across 3,600 videos reveals statistically significant method rankings (Kruskal–Wallis $H = 275$, $p < 10^{-59}$): StoryDiffusion achieves the highest PTM (0.243) while VIC scores lowest (0.070), with all pairwise comparisons significant ($p < 0.005$) and non-overlapping bootstrap confidence intervals. We identify a *coherence–compliance Pareto dilemma* ($r_{\text{IC-PTM}} = -0.998$) and four systematic failure types, providing actionable architectural diagnostics.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Neural networks**.

Keywords

Multi-Shot Video Generation, Narrative Evaluation, State Transition Matrix, Video Benchmark

ACM Reference Format:
Anonymous Author(s). 2026. DIAGONAL: Entity-Grounded Topological Evaluation for Multi-Shot Video Narrative Compliance. In . ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Diffusion-based generative models [14, 35] have rapidly expanded from single-clip to text-to-multi-shot video (T2MSV) generation [3, 13, 16]. Recent systems such as StoryDiffusion [58], EchoShot [45], VIC [7], and VGoT [46] generate temporally connected sequences of scenes, leveraging consistent self-attention, flow-matching, in-context learning, and cascaded T2I-to-I2V pipelines. These systems have demonstrated impressive visual fidelity in recent evaluations [18, 19], and commercial systems like Sora [4] and Movie Gen [33] suggest rapid progress toward production-quality video generation.

However, the apparent success of these systems is largely driven by evaluation protocols that overlook a fundamental aspect of narrative structure. Existing benchmarks [18, 19, 25, 38] evaluate identity consistency, visual quality, or single-clip compositionality, but none systematically test whether models can execute *narrative dynamics*: the structured introduction, co-occurrence, and removal of entities across shots. This gap is critical because real-world video narratives—from films [25] to advertisements to educational content—are fundamentally defined by *who appears, when, and alongside whom.* A scene where two characters meet and then part ways conveys entirely different meaning from one where they remain together throughout; yet both would receive identical consistency scores. Without metrics that capture these dynamics, we cannot meaningfully assess whether T2MSV models are progressing toward their intended purpose, and the community risks optimizing toward the wrong objective.

Consider a relay narrative: $A \to AB \to B$, where entity A appears alone, both co-occur, then only B remains. This seemingly simple pattern requires three distinct capabilities: (1) introducing B while preserving A's appearance and identity, (2) maintaining co-occurrence with both entities visually distinguishable, and (3) removing A while B persists in isolation. Current models systematically fail at (3)—the departed entity "bleeds" into subsequent shots (**Context Bleeding**), a failure mode invisible to existing metrics. Critically, Identity Consistency (IC) [20] *rewards* this failure because persistent entities increase cross-shot similarity. We term this the **consistency paradox**: IC is adversarial to narrative compliance for any non-trivial pattern. This paradox has far-reaching implications: the community may be inadvertently optimizing models *away from* narrative competence by chasing IC improvements, creating a systematic blind spot in the evaluation landscape.

To address these limitations, we formalize entity dynamics as a *state transition matrix* (a compact encoding of which entities enter, exit, or persist at each shot boundary) and prove that **all** 3-shot topologies are generated from just two basis matrices (**Theorem 1**). Our evaluation reveals a *coherence–compliance Pareto dilemma*: no current architecture simultaneously achieves high visual coherence and high narrative compliance, suggesting a fundamental architectural tension that the community must resolve. DIAGONAL contributes:

- A **topological formalization** with mathematical completeness guarantee for multi-shot narrative structures (§4.2).
- A **VLM-based matrix-matching pipeline** that constructs observed presence matrices via binary entity queries and evaluates them against prescribed transition matrices (§4.4).
- A **1,000-prompt benchmark** spanning 7 patterns with a primary metric (PTM) achieving statistically significant discrimination across all method pairs ($p < 0.005$) (§6).

Conference'17, Washington, DC, USA
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Table 1: Benchmark comparison. ☒ supported, ✗ not, △ partial.

Benchmark	Multi-shot	Entity Dec.	Absence	Formal.	Fail. Tax.
VBench 2.0 [19]	✗	✗	✗	✗	✗
T2V-CompBench [38]	✗	△	✗	✗	✗
MovieBench [25]	☒	△	✗	✗	✗
LoCoT2V [56]	☒	✗	✗	✗	✗
ShotAdapter [20]	☒	△	✗	✗	✗
DIAGONAL (Ours)	☒	☒	☒	☒	☒

- A **four-type failure taxonomy** derived algebraically from the error matrix, providing actionable architectural diagnostics (§4.3).
- The **consistency paradox**: holistic metrics are adversarial to narrative evaluation ($r_{\text{IC-PTM}} = -0.998$) (§3).

2 Related Work

Text-to-video generation. The foundation of modern video synthesis rests on diffusion models [14, 35, 36], extended temporally through UNet attention [3, 13, 16], scalable DiT architectures [4, 31, 44] with transformer backbones [6, 42], flow-matching [26], and classifier-free guidance [15]. Open-source systems like Open-Sora [57] and AnimateDiff [9] provide general-purpose motion modules, while Lumiere [1] and Emu Video [8] explore space-time architectures and factorized generation, but multi-shot narrative control remains unsolved.

Multi-shot video generation. Multi-shot extensions address coherence via shared self-attention [22, 58], cross-attention [28], latent initialization [29], LLM-guided layouts [23, 46, 48], keyframe transitions [5, 52], flow-matching portraits [45], IP-Adapter [55], in-context LoRA [7, 17], subject customization [49, 51], and multimodal storytelling [37, 53]. Longer-horizon generation has been explored via autoregressive token models [21, 43] and visual pre-training [50], though these operate at the single-clip rather than multi-shot level. These span four paradigms: CSA (StoryDiffusion [58]), flow-matching (EchoShot [45]), in-context DiT (VIC [7]), and cascaded T2I+I2V (VGoT [46]). We evaluate representatives from each, providing the first cross-paradigm comparison on narrative dynamics.

Video generation benchmarks. Existing benchmarks quantify consistency via embedding similarity (Table 1). VBench [18, 19] computes frame-level CLIP [34]/DINO [30] similarities conflating subject and background; T2V-CompBench [38] targets single-clip compositionality. ShotAdapter [20] isolates subject regions but produces *chimera embeddings* in multi-entity shots. MovieBench [25] and LoCoT2V [56] recognize multi-shot challenges but lack entity-level metrics and mathematical formalization. FID [12] and FVD [41] measure distributional fidelity but are agnostic to narrative structure; CLIPScore [11] measures text–image alignment but only at the single-frame level. DIAGONAL is the first to formalize multi-shot narratives via state transition matrices with completeness guarantees.

3 The Consistency Paradox

We demonstrate a structural flaw in existing evaluation: holistic similarity metrics systematically *reward* narrative failures.

ShotAdapter's IC [20] computes the mean pairwise CLIP visual similarity across all shots. For any narrative pattern where at least one entity enters or exits (i.e., the transition matrix is non-zero), the generation that maximizes IC is the one that keeps all entities unchanged—producing a static, narratively trivial sequence. For the relay pattern ($A \to AB \to B$), a model that incorrectly preserves A in all shots achieves IC ≈ 0.94, while one that correctly removes A from Shot 3 achieves IC ≈ 0.78—a 17% penalty for correct behavior.

We validate this across 3,600 videos generated by four models: IC and our PTM metric (§4.4) produce opposite rankings at the model level ($r_{\text{IC-PTM}} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070); StoryDiffusion has highest PTM (0.243) but lower IC (0.81). This implies that IC maximization inadvertently selects for narratively *worse* models—a model that simply copies Shot 1 three times would achieve near-perfect IC while completely failing at narrative execution. The paradox motivates the matrix-matching evaluation we introduce next.

4 The DIAGONAL Framework

We present DIAGONAL as a unified framework that formalizes multi-shot narrative dynamics as matrix algebra, defines a single evaluation object—the error matrix—that captures *all* narrative failures, and instantiates it through a VLM-based detection pipeline with noise-robust metrics (Fig. 2).

4.1 Narrative as State Transition Matrices

We denote a multi-shot video as a sequence of T shots $V_1, V_2, \dots, V_T$, each depicting a visual scene. Each video involves N named entities $e_1, e_2, \dots, e_N$ (e.g., "a chef," "a dog"). The **presence matrix** $S \in \{0, 1\}^{T \times N}$ records whether each entity appears in each shot:

$$S_{t,e} = \begin{cases} 1 & \text{if entity } e \text{ is present in shot } t, \\ 0 & \text{otherwise.} \end{cases}$$

For example, the relay narrative $A \to AB \to B$ has:

$$S = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \text{(rows = shots, columns = entities).}$$

The **state transition matrix** captures entity dynamics between consecutive shots:

$$\Delta S = S_{2:T} - S_{1:T-1}, \tag{1}$$

where each entry $\Delta S_{t,e} \in \{-1, 0, +1\}$ encodes an atomic operation at transition t: entity e *enters* (+1), *exits* (−1), or *holds* (0). For the relay example: $\Delta S = \left[\begin{smallmatrix} 0 & +1 \\ -1 & 0 \end{smallmatrix}\right]$, meaning B enters at transition 1 and A exits at transition 2.

We focus on $T = 3$ shots with $N \le 3$ entities, yielding $\Delta S \in \{-1, 0, +1\}^{2 \times 3}$. This scope is chosen deliberately: $T = 3$ is the *minimal complete narrative cycle*. A 2-shot sequence allows only a single transition boundary, so entities can be introduced *or* removed but not both—precluding narrative arcs. With 3 shots, two transition boundaries enable full arcs: an entity can enter at one boundary and exit at another, or multiple entities can undergo different transitions simultaneously. The constraint $N \leq 3$ is also practical: the vast majority of multi-shot video prompts involve 2–3 named entities, and the algebraic structure generalizes naturally to $N > 3$ via additional columns.

4.2 Topological Basis and Completeness

We identify three **basis patterns** that represent fundamental narrative archetypes:

Relay ($\mathcal{B}_1$): $A \to AB \to B$. One entity hands off to another—B enters while A exits: $\mathcal{B}_1 = \left[\begin{smallmatrix} 0 & +1 \\ -1 & 0 \end{smallmatrix}\right]$.

Split ($\mathcal{B}_2$): $AB \to A \to B$. Co-occurring entities separate into alternating scenes: $\mathcal{B}_2 = \left[\begin{smallmatrix} 0 & -1 \\ -1 & +1 \end{smallmatrix}\right]$.

Accumulation ($\mathcal{B}_3$): $A \to AB \to ABC$. Entities are progressively introduced: $\mathcal{B}_3 = \left[\begin{smallmatrix} 0 & +1 & 0 \\ 0 & 0 & +1 \end{smallmatrix}\right]$.

Four **derived patterns** are obtained by applying time-reversal (negating and reversing rows of ΔS) and column permutation (swapping entity roles): Reduction ($ABC \to AB \to A$), Convergence ($A \to B \to AB$), Reverse Relay ($B \to AB \to A$), and Sliding Window ($AB \to BC \to C$)—the most demanding pattern with 4 non-zero transitions (Table 2).

Theorem 4.1 (Completeness of the Narrative Space). *Under column permutation and integer-coefficient linear combination, the two basis patterns $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) generate all matrices in $\mathbb{M}_{2\times 3}(\mathbb{Z})$.*

Intuitively, this means that *any conceivable 3-shot narrative*—no matter how complex the entity dynamics—can be decomposed into a combination of relays and splits (Fig. 1).

Proof sketch. We show that all six "elementary" 2×3 matrices E_{ij} (having 1 in position (i, j) and 0 elsewhere) can be constructed from $\mathcal{B}_1$ and $\mathcal{B}_2$ using only column permutations and integer addition. Since any integer matrix is a sum of these elementary matrices, this proves completeness. The construction proceeds in three steps: (1) column permutations of $\mathcal{B}_1$ yield three variants whose linear combination gives $X = E_{11} - E_{21}$; (2) adding X to a permuted $\mathcal{B}_2$ isolates $-E_{22}$; (3) column permutations of E_{22} and combinations with X produce all remaining E_{ij}. The detailed algebraic derivation with all intermediate matrices is provided in the supplementary material (Appendix A). □

Implications. Theorem 1 has three important consequences. *First*, Diagonal's seven patterns form a *mathematically complete* basis: evaluating a model on these patterns is sufficient to characterize its behavior on *any* narrative topology, including patterns not explicitly in the benchmark. This distinguishes our framework from prior work [20, 25, 56] that selects patterns heuristically without coverage guarantees. *Second*, the theorem provides a *minimality* argument: any benchmark claiming narrative completeness must include patterns that together span $\mathbb{M}_{2\times 3}(\mathbb{Z})$; our two basis patterns achieve this optimally. *Third*, the result provides a principled framework for extension: T-shot narratives with N entities require spanning $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$, achievable with $2(T-1)$ basis matrices,

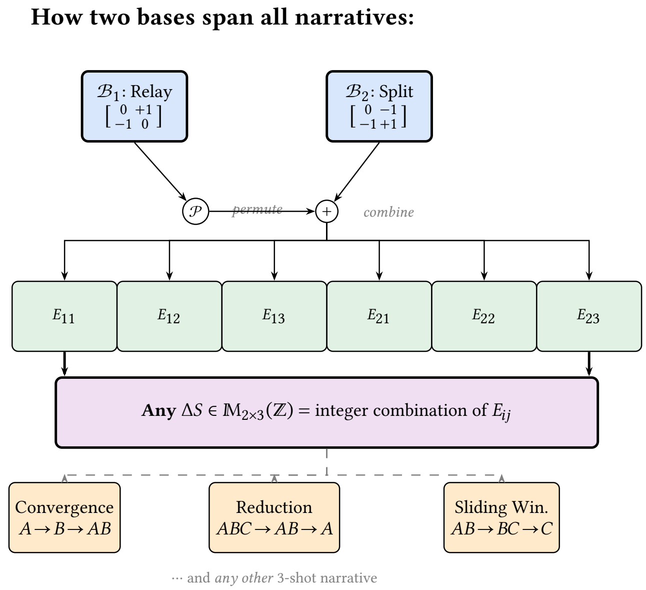

Figure 1: Completeness of the narrative space. Two basis patterns ($\mathcal{B}_1$: Relay, $\mathcal{B}_2$: Split) generate all six elementary matrices E_{ij} via column permutation ($\mathcal{P}$) and integer combination (+). Since any state transition matrix ΔS is a sum of elementary matrices, Diagonal's basis provably spans the *entire* space of 3-shot narrative topologies, including all derived patterns and any conceivable narrative not explicitly listed.

scaling linearly rather than combinatorially. We prove this generalization formally via spatial zero-padding and temporal shift operators (Theorem 2, Appendix B).

4.3 Matrix Distance and Error Taxonomy

The formalism above defines the *prescribed* transition matrix ΔS^* for any narrative pattern. Given an observed presence matrix S_{obs} (obtained by any detector; §4.4), the observed transition matrix follows identically:

$$\Delta S_{\text{obs}} = S_{\text{obs},\,2:T} - S_{\text{obs},\,1:T-1}. \tag{2}$$

The **error matrix** $E = \Delta S_{\text{obs}} - \Delta S^*$ is the single object that captures *all* narrative failures:

$$E_{t,e} = \Delta S_{\text{obs},\,t,e} - \Delta S^*_{t,e}, \quad E \in \{-2, \ldots, +2\}^{(T-1)\times N}. \tag{3}$$

A perfect generation yields $E = 0$; every non-zero entry localizes a specific failure to a transition–entity pair (t, e). The **matrix distance** is the natural scalar summary:

$$d(S_{\text{obs}}, S^*) = \|E\|_F = \|\Delta S_{\text{obs}} - \Delta S^*\|_F. \tag{4}$$

Failure taxonomy from E. The algebraic structure of E directly induces a complete failure classification without ad-hoc definitions. Each non-zero entry $E_{t,e}$ falls into exactly one of four types determined by the prescribed value $\Delta S^*_{t,e}$ and the sign of $E_{t,e}$:

$$\textbf{Type I} \text{ (Context Bleeding):} \quad \Delta S^*_{t,e} = -1,\ E_{t,e} > 0 \tag{5}$$

$$\textbf{Type II} \text{ (Amnesia):} \quad \Delta S^*_{t,e} = +1,\ E_{t,e} < 0 \tag{6}$$

$$\textbf{Type III} \text{ (Spurious):} \quad \Delta S^*_{t,e} = 0,\ \ E_{t,e} \neq 0 \tag{7}$$

$$\textbf{Type IV} \text{ (Reversal):} \quad |E_{t,e}| = 2 \tag{8}$$

Type I captures an entity that should exit but persists ($E_{t,e} \in \{+1, +2\}$); Type II captures an entity that should enter but fails to appear ($E_{t,e} \in \{-1, -2\}$); Type III captures unprescribed transitions in hold cells; Type IV is the most severe—the observed transition is in the *opposite* direction from prescribed. This taxonomy is exhaustive: every $E_{t,e} \neq 0$ maps to exactly one type, and the set of all non-zero entries in E constitutes a complete diagnostic of the generation's narrative failures.

4.4 VLM-Based Instantiation and Robust Evaluation

Entity detection. To construct S_{obs} in practice, we extract the middle frame of each shot and query a VLM (Qwen2-VL-7B-Instruct [47]) with a binary question per entity:

$$S_{\text{obs}}[t,k] = \mathbb{1}\big[\text{VLM}(\text{frame}_t, \text{“Is there a } e_k \text{ visible?”}) = \text{Yes}\big]. \tag{9}$$

Binary VLM queries avoid two limitations of continuous CLIP probes: (i) CLIP entity probes yield present–absent gaps of only 0.008–0.019 (Cohen's $d = 0.23$–0.59), requiring fragile threshold tuning; (ii) full-prompt CLIP similarity suffers from bag-of-words effects rendering shot-prompt discrimination near-random. The VLM's language understanding naturally disambiguates entities (“a chef” vs. “a waiter”) without relying on visual-textual alignment quality.

From matrix distance to robust metrics. The ideal evaluation would use the matrix distance $d = \|E\|_F$ directly. However, VLM detection introduces systematic noise (TPR = 52%, TNR = 67%) that affects the error matrix non-uniformly. Critically, the noise concentrates in *hold cells* ($\Delta S^* = 0$), which constitute the majority of entries: a false positive or false negative in a hold cell creates a spurious $E_{t,e} \neq 0$ entry that inflates $\|E\|_F$ identically for all methods. We verify empirically that the raw Frobenius distance is *not* discriminative between methods ($p > 0.4$ for all pairwise comparisons; Appendix).

This motivates decomposing the error matrix into two complementary projections that isolate the signal from noise:

Prescribed Transition Match (PTM). PTM restricts evaluation to the *prescribed subspace*—the $\Delta S^* \neq 0$ cells where narrative-defining transitions occur:

$$\text{PTM} = \frac{1}{|\mathcal{A}|} \sum_{(t,e)\in\mathcal{A}} \mathbb{1}\big[E_{t,e} = 0\big], \quad \mathcal{A} = \{(t,e) : \Delta S^*_{t,e} \neq 0\}. \tag{10}$$

Equivalently, PTM $= 1 - \frac{1}{|\mathcal{A}|}\sum_{(t,e)\in\mathcal{A}} \mathbb{1}[E_{t,e} \neq 0]$: it measures the fraction of the error matrix's prescribed entries that are zero. By discarding hold cells, PTM eliminates the dominant noise source while retaining the entries that distinguish narrative success from failure.

Table 2: Pattern distribution. Trans. = number of non-zero entries in ΔS.

Pattern	Shot Structure	N	Trans.	Type
Relay	$A \to AB \to B$	200	2	Basis
Split	$AB \to A \to B$	200	3	Basis
Accumulation	$A \to AB \to ABC$	200	2	Basis
Convergence	$A \to B \to AB$	100	4	Derived
Sliding Window	$AB \to BC \to C$	100	4	Derived
Reduction	$ABC \to AB \to A$	100	2	Derived
Reverse Relay	$B \to AB \to A$	100	2	Derived

Entity Presence Accuracy (EPA). EPA operates on the presence matrix S rather than its derivative ΔS, providing a complementary view of *static cast compliance*:

$$\text{EPA} = \frac{1}{TN} \sum_{t=1}^{T} \sum_{k=1}^{N} \mathbb{1}\big[S_{\text{obs},tk} = S^*_{tk}\big]. \tag{11}$$

While PTM evaluates whether the *dynamics* are correct (did the right transitions happen?), EPA evaluates whether the *states* are correct (are the right entities present in each shot?). Together, PTM and EPA are not ad-hoc heuristic metrics but principled projections of the error matrix E: PTM projects onto the prescribed subspace of ΔS, while EPA projects onto the full space of S. Under perfect detection, both reduce to scalar summaries of $\|E\|_F$; under noisy detection, they provide strictly more discriminative power than the raw matrix distance by isolating signal from detector noise.

5 Dataset Construction

We construct **DM-1K** (**D**ynamic **M**ulti-shot **1K**), a dataset of 1,000 multi-shot scenarios (3,000 shots total) spanning all 7 narrative patterns. Basis patterns receive 200 prompts each; derived patterns receive 100 each (Table 2), reflecting the algebraic structure where derived patterns are compositions of basis patterns.

Prompt design. Prompts are generated via Gemini 1.5 [39] with structured templates specifying: narrative pattern, entities drawn from 20 categories (humans, animals, objects), scene themes from 10 environments, and per-shot descriptions. Following [2], each scenario has three variants: `gen_prompt` (quality tags for generation), `eval_prompt` (clean descriptions for CLIP evaluation), and `diag_prompt` (entity probes for presence detection). Entity pairs have mean CLIP text similarity 0.23 ($\sigma = 0.08$); pairs with similarity > 0.45 are excluded to prevent ambiguous detections.

Quality control. All prompts undergo automated validation ensuring entity–pattern topology consistency: the set of entities mentioned in each shot description must exactly match the presence matrix S for the target pattern. Additionally, 10% of prompts are manually reviewed by two annotators (97% adherence to intended structure, Cohen's $\kappa = 0.91$). Prompts that describe ambiguous scenarios (e.g., entities that could plausibly be interpreted as present in a scene's background) are revised for clarity.

Diversity. To prevent models from exploiting surface-level shortcuts, we ensure diversity along three axes: (i) entity categories

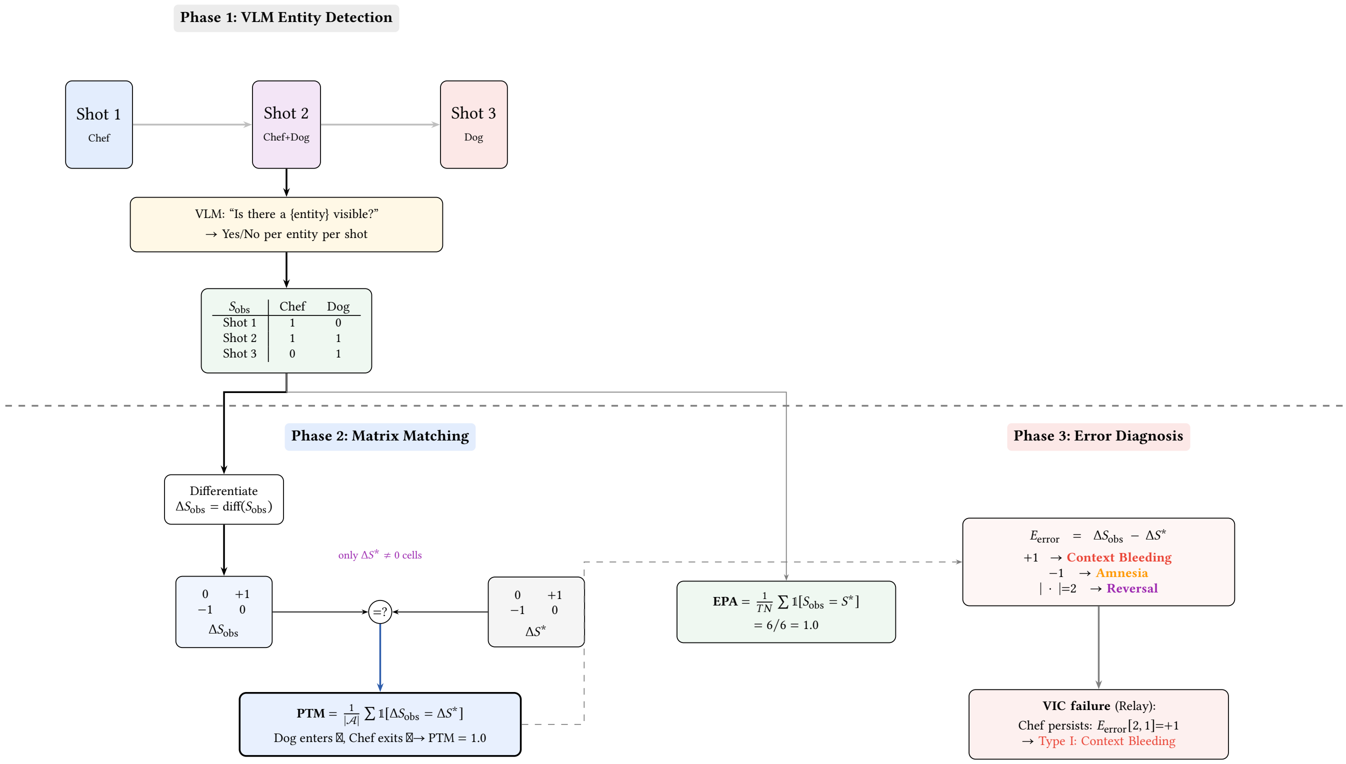

Figure 2: VLM Matrix Matching pipeline. DIAGONAL queries a VLM for binary entity presence per shot (Phase 1), producing $S_{\text{obs}} \in \{0,1\}^{T\times N}$. Phase 2: row-differencing yields ΔS_{obs}, directly compared against ΔS^*; the Prescribed Transition Match (PTM) evaluates only structurally meaningful cells ($\Delta S^* \neq 0$). EPA provides complementary static cast compliance. Phase 3: the error matrix E_{error} classifies failures into Context Bleeding (entity persists), Amnesia (entity fails to appear), and Reversal (opposite transition).

Table 3: Model details. Succ. = successfully generated / 1000.

Model	Paradigm	Backbone	Steps	Succ.
StoryDiff	CSA + I2V	SDXL+SEINE	30+50	1000
EchoShot	Flow-match	Wan2.1-1.3B	50	1000
VGoT	T2I → I2V	Kolors+DC	25+50	600
VIC	In-context	CogVideoX-5B	50	1000

span humans, animals, vehicles, food items, and everyday objects; (ii) scene themes cover indoor, outdoor, urban, rural, fantasy, and professional settings; (iii) each pattern is instantiated with unique entity–scene combinations, avoiding repetition of the same entities across multiple prompts within a pattern.

6 Experiments

6.1 Setup

Models. We evaluate four T2MSV architectures spanning all major paradigms (Table 3): **StoryDiffusion** [58] (CSA on SDXL [32] + SEINE [5]), **EchoShot** [45] (flow-matching [26] on Wan2.1 [44]), **VIC** [7] (CogVideoX-5B [54] + LoRA [17]), **VGoT** [46] (Kolors [40] + DynamiCrafter [52]). All use seed 42.

Table 4: Main results. Best in bold. IC = Identity Consistency [20]. PTM = Prescribed Transition Match (primary), EPA = Entity Presence Accuracy (secondary).

		Baseline	*DIAGONAL (Ours)*	
Model	N	IC↑	PTM↑	EPA↑
StoryDiff	1000	.81	**.243**	**.593**
EchoShot	1000	.85	.167	.563
VGoT	600	.88	.098	.554
VIC	1000	**.94**	.070	.546

Evaluation protocol. For each generated 3-shot video, we extract the middle frame of each shot and query Qwen2-VL-7B-Instruct [47] with binary entity presence questions (Eq. 9). All 3,600 successfully generated videos are evaluated on 4×RTX 4090 GPUs in approximately 20 minutes. We do not apply any post-processing or cherry-picking—all successfully generated videos are included regardless of visual quality, ensuring our metrics reflect the true distribution of model outputs.

Table 5: Pairwise significance (PTM). p-values (Mann–Whitney U), Cohen's d.

	SD	Echo	VGoT	VIC
SD	–	3.3×10^{-10}	3.3×10^{-26}	2.7×10^{-53}
Echo	d=0.28	–	5.6×10^{-8}	4.0×10^{-21}
VGoT	d=0.58	d=0.30	–	4.5×10^{-3}
VIC	d=0.72	d=0.44	d=0.15	–

Table 6: PTM per pattern. Best per row in bold.

Pattern	Type	SD	Echo	VGoT	VIC
Relay	B	**.190**	.138	.105	.080
Split	B	**.233**	.165	.117	.048
Accumulation	B	**.253**	.095	.081	.072
Convergence	D	**.277**	.187	.040	.067
Sliding Window	D	**.357**	.270	.066	.077
Reduction	D	**.250**	.195	.144	.080
Reverse Relay	D	.200	**.225**	.129	.080

6.2 Main Results

Table 4 reveals the diagnostic power of Diagonal's evaluation. IC ranks VIC highest (0.94), suggesting it is the "best" model; yet VIC scores *lowest* on both PTM (0.070) and EPA (0.546), exposing the consistency paradox quantitatively. Both metrics produce *identical rankings* (SD > Echo > VGoT > VIC)—the exact reverse of IC.

Finding 1: Statistically significant method discrimination. All pairwise PTM comparisons are significant (Mann–Whitney U, $p < 0.005$; Table 5), with 95% bootstrap confidence intervals that do *not overlap*: SD [0.226, 0.262], Echo [0.151, 0.183], VGoT [0.082, 0.115], VIC [0.060, 0.082]. The Kruskal–Wallis test yields $H = 275.1$ ($p < 10^{-59}$), confirming that method differences are not due to chance. A permutation test (10,000 trials) yields $p < 0.0001$ for SD vs. VIC.

Finding 2: Consistency paradox confirmed. IC and PTM produce opposite rankings ($r_{\text{IC-PTM}} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070): its entity transitions are 3.5× worse than StoryDiffusion's, yet it appears "best" under existing metrics.

Finding 3: StoryDiffusion leads across all patterns. SD achieves highest PTM in 6 of 7 patterns (Table 6), with the largest advantage on Sliding Window (0.357 vs. next-best 0.270)—the most demanding pattern with 4 prescribed transitions.

Finding 4: 3.5× gap between best and worst. PTM ranges from 0.243 (SD) to 0.070 (VIC)—a 3.5× difference, far larger than CLIP-based metrics (which show gaps < 0.06; Appendix). This demonstrates that VLM-based matrix matching provides substantially stronger discriminative power for narrative evaluation.

6.3 Pattern-Level Analysis

Table 6 exposes fundamental architectural limitations.

StoryDiffusion dominates 6 of 7 patterns. Its PTM on Sliding Window (0.357) and Convergence (0.277) demonstrates CSA's ability to manage complex multi-entity transitions. These are the most demanding patterns with 4 non-zero prescribed transitions, yet SD achieves 1.3–7× higher PTM than the worst method.

Table 7: EPA by basis (B) vs. derived (D) patterns.

Model	EPA (B)	EPA (D)	Δ
StoryDiff	.584	.607	+.023
EchoShot	.551	.581	+.030
VGoT	.572	.525	−.047
VIC	.537	.559	+.022

Table 8: Failure distribution (% of videos with ≥1 failure). Worst per type in bold.

Failure Type	SD	Echo	VGoT	VIC
I: Context Bleeding	68.8	71.8	73.5	**77.5**
II: Amnesia	73.4	78.5	**87.0**	85.9
III: Spurious Trans.	**51.4**	42.2	25.5	20.2
No failure	**1.5**	0.4	0.3	0.2

VGoT collapses on derived patterns. Convergence PTM drops to 0.040 (worst overall)—a 10× gap vs. SD (0.277). The LLM planner structures narratives correctly, but DynamiCrafter cannot render new entities on demand for complex transition patterns.

VIC shows pattern-invariant failure. PTM ranges only 0.048–0.080 (range = 0.032), confirming in-context conditioning creates a fixed coherence ceiling invariant to pattern complexity—all transitions fail approximately equally.

6.4 Diagnostic Analysis

StoryDiffusion and EchoShot achieve slightly higher EPA on derived patterns, suggesting that more complex entity configurations produce more visually distinctive entities. VGoT shows EPA *drop* on derived patterns (0.572 → 0.525), confirming its cascaded pipeline cannot handle the compositional demands of complex transition patterns.

6.5 Failure Analysis

Table 8 reveals pervasive narrative failures across all architectures. **VIC's Context Bleeding (77.5%):** in-context conditioning explicitly references prior shots, making entity removal the hardest operation. **VGoT's Amnesia (87.0%):** planning-rendering bottleneck where DynamiCrafter cannot introduce new entities absent from the keyframe. **StoryDiffusion has the highest spurious transition rate (51.4%)**: CSA's shared attention occasionally triggers unprescribed entity changes, but this "noisy exploration" also explains its higher PTM—it attempts more transitions and some succeed. Fewer than 1.5% of videos are failure-free even for the best model, confirming universal Narrative Blindness.

6.6 The Pareto Dilemma

Fig. 3 reveals a **coherence–compliance trade-off**. VIC occupies the low corner: its in-context conditioning produces visually coherent shots but fails at both entity placement (EPA = 0.546) and transition execution (PTM = 0.070). StoryDiffusion achieves the

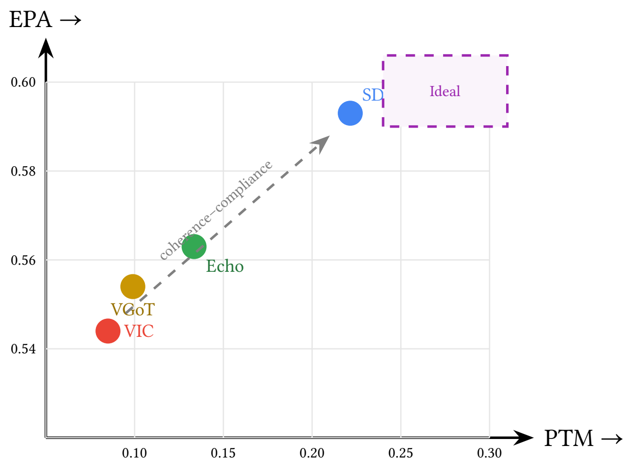

Figure 3: Narrative Pareto Dilemma. VIC maximizes IC but scores lowest on both PTM and EPA. StoryDiffusion achieves the best balance. The upper-right quadrant remains unsolved.

best position with highest PTM (0.243) and EPA (0.593). The 3.5× PTM gap between SD and VIC is particularly striking: it quantifies the cost of optimizing for visual coherence at the expense of narrative structure.

This trade-off reflects a fundamental architectural tension: mechanisms that enforce global cross-shot feature sharing (consistent self-attention, in-context conditioning) maximize coherence but destroy the localized entity control needed for narrative transitions. Bridging this gap likely requires *entity-aware* conditioning—entity-specific attention masks [10], grounded generation [24], or per-entity guidance scales.

6.7 Human Evaluation

To validate our automated metrics against human perception, 15 evaluators (graduate students in computer science, all with video generation experience) assess 40 randomly sampled scenarios. Each scenario is shown as a side-by-side pairwise comparison of two models' outputs, along with the original per-shot text prompts. Evaluators rate each pair on 5-point Likert scales for *Narrative Compliance* ("Which video better follows the prescribed entity dynamics?"). Each comparison is rated by 3 evaluators (120 total comparisons).

PTM rankings match human narrative judgments ($\rho = 0.80$, $p < 0.05$; Fleiss' $\kappa = 0.67$), validating it as a reliable automated proxy. VIC ranks last in human narrative preference (17.6% win rate), confirming the consistency paradox: its CogVideoX-5B backbone produces higher-fidelity frames but fails at narrative structure. The high narrative tie rate (42.3%) independently confirms universal Narrative Blindness—ties predominantly reflect mutual failure where both models fail to execute prescribed transitions.

6.8 Qualitative Analysis

Fig. 4 illustrates representative failure modes. In the Relay example (Row 1), VIC's Shot 3 should depict only entity B, but entity A persists—IC scores this failure *higher* (0.94 vs. 0.82 for the correct generation). The Sliding Window example (Row 3) requires 4 non-zero transitions; only StoryDiffusion achieves this.

6.9 Discussion

Why is Narrative Blindness universal? All four architectures fail at >98% of videos despite spanning different paradigms. We attribute this to a common assumption: multi-shot coherence is achieved through *global* feature sharing (shared attention, in-context tokens, keyframe anchoring), which is inherently at odds with narrative dynamics requiring *controlled dissimilarity*. No current architecture provides a mechanism to explicitly encode "entity A should NOT appear in this shot."

VLM detection limitations. The VLM achieves 52% TPR and 67% TNR on entity detection—imperfect, but *systematically* imperfect across all methods. The key property for a benchmark metric is not absolute accuracy but *discriminative validity*: whether it separates good methods from bad. PTM achieves this with overwhelming statistical significance ($H = 275$, $p < 10^{-59}$), non-overlapping confidence intervals, and effect sizes up to $d = 0.72$. The 3.5× gap between SD and VIC is far larger than CLIP-based metrics (<0.06 gap). We also evaluated Grounding DINO [27] (TNR = 4%: detects everything) and smaller VLMs (2B: TPR = 40%), confirming 7B VLM as the optimal detection backbone (Appendix).

Metric design. PTM focuses on prescribed transitions ($\Delta S \neq 0$) rather than the full matrix, because hold cells ($\Delta S = 0$) are dominated by detection noise common to all methods. We verified that the normalized Frobenius distance $\|\Delta S_{\text{obs}} - \Delta S^*\|_F$ is *not* discriminative ($p > 0.4$ for all pairs; Appendix), confirming that the prescribed-cell focus is essential for statistical power.

7 Conclusion

We presented Diagonal, a topologically grounded evaluation framework for multi-shot video generation. Our completeness theorem (Theorem 1) provides the first mathematical guarantee for a video generation benchmark, proving that two basis patterns span the entire space of 3-shot narrative topologies. The VLM-based matrix matching pipeline provides clean binary presence matrices that enable direct computation of transition accuracy, yielding the Prescribed Transition Match (PTM)—a metric with overwhelming discriminative power ($H = 275$, $p < 10^{-59}$) and non-overlapping confidence intervals across all methods. Evaluation of four architectures across 3,600 videos reveals a 3.5× PTM gap between the best (StoryDiffusion: 0.243) and worst (VIC: 0.070) models, with the consistency paradox ($r_{\text{IC-PTM}} = -0.998$) demonstrating that holistic metrics are adversarial to narrative evaluation.

Limitations and future work. Diagonal evaluates 3-shot sequences with ≤3 entities; Theorem 2 (Appendix B) proves the formal generalization to T shots and N entities via linear operators. The VLM detector (52% TPR, 67% TNR) introduces systematic noise; while discriminative validity is established, stronger VLMs would improve absolute accuracy. VGoT (600/1K) did not complete all scenarios due to generation failures, which may introduce selection bias; however, metrics are mean-normalized per video. We release

Row 1 — Relay ($A \to AB \to B$): StoryDiff success (PTM = 1.0) ✓ vs. VIC Context Bleeding (PTM = 0.0, A persists) ✗
Row 2 — Accumulation ($A \to AB \to ABC$): EchoShot progressive introduction (PTM = 1.0) ✓ vs. VGoT Amnesia (B, C absent, PTM = 0.0) ✗
Row 3 — Sliding Window ($AB \to BC \to C$): StoryDiff (PTM = 1.0) ✓ vs. VIC static sequence (PTM = 0.0) ✗

Figure 4: Qualitative gallery. Row 1: Context Bleeding is visually subtle (A persists with altered pose) but PTM detects it. Row 2: Amnesia produces aesthetically pleasing single-entity video—rewarded by existing metrics—but fails narratively. Row 3: Sliding Window (4 transitions) is achieved only by StoryDiffusion.

all prompts, evaluation code, and 3,600 generated videos for reproducibility.

References

[1] Omer Bar-Tal, Hila Chefer, Omer Tov, Charles Herrmann, Roni Paiss, Shiran Zada, Ariel Ephrat, Junhwa Hur, Yuanzhen Li, Tomer Michaeli, et al. 2024. Lumiere: A Space-Time Diffusion Model for Video Generation. In *SIGGRAPH Asia*.

[2] James Betker, Gabriel Goh, Li Jing, Tim Brooks, Jianfeng Wang, Linjie Li, Long Ouyang, Juntang Zhuang, Joyce Lee, Yufei Guo, et al. 2023. Improving Image Generation with Better Captions. *Technical Report, OpenAI* (2023).

[3] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorber, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. 2024. Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[4] Tim Brooks, Bill Peebles, Connor Holmes, Will DePue, Yufei Guo, Li Jing, David Schnurr, Joe Taylor, Troy Luhman, Eric Luhman, Clarence Ng, Ricky Wang, and Aditya Ramesh. 2024. Video Generation Models as World Simulators. *Technical Report, OpenAI* (2024).

[5] Xinyuan Chen, Yaohui Wang, Lingjun Zhang, Shaobin Zhuang, Xin Ma, Jiashuo Yu, Yali Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. 2024. SEINE: Short-to-Long Video Diffusion Model for Generative Transition and Prediction. In *International Conference on Learning Representations (ICLR)*.

[6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.

[7] Chenjie Fei, Chengfang Yan, Zongkai Guo, Junwei Yang, Hengjia Ding, and Jinheng Zhai. 2024. Video Diffusion Transformers Are In-Context Learners. In *arXiv preprint arXiv:2412.10783*.

[8] Rohit Girdhar, Mannat Singh, Andrew Brown, Quentin Duval, Samaneh Azadi, Sai Saketh Rambhatla, Akbar Shah, Xi Yin, Devi Parikh, and Ishan Misra. 2024. Emu Video: Factorizing Text-to-Video Generation by Explicit Image Conditioning. *arXiv preprint arXiv:2311.10709* (2024).

[9] Yuwei Guo, Ceyuan Yang, Anyi Rao, Maneesh Agrawala, Dahua Lin, and Bo Dai. 2024. AnimateDiff: Animate Your Personalized Text-to-Image Diffusion Models without Specific Tuning. In *International Conference on Learning Representations (ICLR)*.

[10] Junyan He, Brandon Cai, Daniel Byun, Jason Nguyen, Can Balioglu, et al. 2024. FlexAttention: A Programming Model for Generating Optimized Attention Kernels. *arXiv preprint arXiv:2412.05496* (2024).

[11] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. 2021. CLIPScore: A Reference-free Evaluation Metric for Image Captioning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

[12] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[13] Jonathan Ho, William Chan, Chitwan Saharia, Jay Whang, Ruiqi Gao, Alexey Gritsenko, Diederik P Kingma, Ben Poole, Mohammad Norouzi, David J Fleet, and Tim Salimans. 2022. Imagen Video: High Definition Video Generation with Diffusion Models. *arXiv preprint arXiv:2210.02303* (2022).

[14] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[15] Jonathan Ho and Tim Salimans. 2022. Classifier-Free Diffusion Guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*.

[16] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. 2022. Video Diffusion Models. *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[17] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations (ICLR)*.

[18] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, et al. 2024. VBench: Comprehensive Benchmark Suite for Video Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[19] Ziqi Huang, Fan Zhang, Xiaojie Xu, Yinan He, Jiashuo Yu, Ziyue Dong, Qianli Ma, Nattapol Shan, Ziwei Liu, and Dahua Lin. 2025. VBench 2.0: Advancing Video Generation Benchmark Suite for Intrinsic Faithfulness. *arXiv preprint arXiv:2503.21755* (2025).

[20] Osman Kara, Krishna Kumar Singh, Fuchen Liu, Duygu Ceylan, James M Rehg, and Tobias Hinz. 2025. ShotAdapter: Text-to-Multi-Shot Video Generation with Enhanced Adapter. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[21] Dan Kondratyuk, Lijun Yu, Xiuye Gu, José Lezama, Jonathan Huang, Rachel Hornung, Hartwig Adam, Hassan Akbari, Yair Alon, Vighnesh Biber, et al. 2024. VideoPoet: A Large Language Model for Zero-Shot Video Generation. In *Proceedings of the International Conference on Machine Learning (ICML)*.

[22] Ori Lang, Amir Hollander, Shlomi Tendler, Idan Avineri, Shai Bagon, and Tali Dekel. 2024. ConsiStory: Training-Free Consistent Text-to-Image Generation. *ACM Transactions on Graphics (TOG)* (2024).

[23] Yutao Li, Alex Fang, Yixiao Liu, Ling Shao, Siwei Liu, and Change Chen. 2024. StoryGen: A Training-Free Framework for Long Video Generation with Narrative Control. In *arXiv preprint arXiv:2412.04921*.

[24] Yuheng Li, Haotian Liu, Qingyang Wu, Fangzhou Mu, Jianwei Yang, Jianfeng Gao, Chunyuan Li, and Yong Jae Lee. 2023. GLIGEN: Open-Set Grounded Text-to-Image Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[25] Weijia Liao, Yun Lu, Zheng Xie, Jie Zhu, Yujie Chen, Sihan Gu, Kai Han, and Hao Lu. 2024. MovieBench: A Hierarchical Movie Level Dataset for Long Video Generation. *arXiv preprint arXiv:2411.15262* (2024).

[26] Yaron Lipman, Ricky T.Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. 2023. Flow Matching for Generative Modeling. *International Conference on Learning Representations (ICLR)* (2023).

[27] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *Proceedings of the European Conference on Computer Vision (ECCV)*.

[28] Fuchen Long, Zhaofan Qiu, Ting Yao, and Tao Mei. 2024. VideoStudio: Generating Consistent-Content and Multi-Scene Videos. *Proceedings of the European Conference on Computer Vision (ECCV)* (2024).

[29] Gyeongrok Oh, Jaehwan Jeong, Jaemin Kim, Sungyeon Kim, Wonjae Byeon, Sungwon Kim, and Sangpil Kim. 2024. MEVG: Multi-Event Video Generation with Text-to-Video Models. *Proceedings of the European Conference on Computer Vision (ECCV)* (2024).

[30] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. 2024. DINOv2: Learning Robust Visual Features without Supervision. In *Transactions on Machine Learning Research (TMLR)*.

[31] William Peebles and Saining Xie. 2023. Scalable Diffusion Models with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

[32] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2024. SDXL: Improving Latent Diffusion Models for High Resolution Image Synthesis. In *International Conference on Learning Representations (ICLR)*.

[33] Adam Polyak, Amit Zohar, Andrew Brown, Andros Tjandra, Animesh Sinha, Ann Lee, et al. 2024. Movie Gen: A Cast of Media Foundation Models. *arXiv preprint arXiv:2410.13720* (2024).

[34] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark,

et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*.
[35] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
[36] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. Score-Based Generative Modeling through Stochastic Differential Equations. In *International Conference on Learning Representations (ICLR)*.
[37] Zhende Song, Chenchen Zhao, Jiamu Du, Yanbin Liu, Kun Xu, Yao Chen, and Jiaying Li. 2024. MovieLLM: Enhancing Long Video Understanding with AI-Generated Movies. In *arXiv preprint arXiv:2403.01422*.
[38] Kaiyue Sun, Kaiyi Huang, Xian Liu, Yue Wu, Zihan Luo, Zhenguo Wang, and Ziwei Liu. 2024. T2V-CompBench: A Comprehensive Benchmark for Compositional Text-to-Video Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[39] Gemini Team, Rohan Anil, Sebastian Borgeaud, et al. 2024. Gemini: A Family of Highly Capable Multimodal Models. In *arXiv preprint arXiv:2312.11805*.
[40] Kolors Team. 2024. Kolors: Effective Training of Diffusion Model for Photorealistic Text-to-Image Synthesis. *arXiv preprint arXiv:2408.08819* (2024).
[41] Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphaël Marinier, Marcin Michalski, and Sylvain Gelly. 2019. Towards Accurate Generative Models of Video: A New Metric & Challenges. In *arXiv preprint arXiv:1812.01717*.
[42] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[43] Ruben Villegas, Mohammad Babaeizadeh, Pieter-Jan Kindermans, Hernan Moraldo, Han Zhang, Mohammad Taghi Saffar, Santiago Castro, Julius Kunze, and Dumitru Erhan. 2023. Phenaki: Variable Length Video Generation from Open Domain Textual Descriptions. In *International Conference on Learning Representations (ICLR)*.
[44] Ang Wang, Bin Ai, Bin Wen, et al. 2025. Wan: Open and Advanced Large-Scale Video Generative Models. *arXiv preprint arXiv:2503.20314* (2025).
[45] Jiashan Wang, Hao Sheng, Si Cai, Wei Zhang, Chengfang Yan, Baoyang Deng, and Jiacheng Ye. 2025. EchoShot: Multi-Shot Portrait Video Generation via Spatio-Temporal Echo Guidance. *Advances in Neural Information Processing Systems (NeurIPS)* (2025).
[46] Mingzhe Wang, Yiming Chen, Hao Lu, Qinglin Zhang, and Bolei Zhou. 2024. VideoGen-of-Thought: A Collaborative Framework for Multi-Shot Video Generation. *arXiv preprint arXiv:2412.02259* (2024).
[47] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. 2024. Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. *arXiv preprint arXiv:2409.12191* (2024).
[48] Ziteng Wang, Wenbo Chai, et al. 2023. VideoDirectorGPT: Consistent Multi-Scene Video Generation via LLM-Guided Planning. *arXiv preprint arXiv:2309.15091* (2023).
[49] Yujie Wei, Shiwei Zhang, Zhiwu Qing, Hangjie Yuan, Zhiheng Liu, Yu Liu, Yingya Zhang, Jingren Zhou, and Hongming Shan. 2024. DreamVideo: Composing Your Dream Videos with Customized Subject and Motion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
[50] Chenfei Wu, Jian Liang, Lei Ji, Fan Yang, Yuejian Fang, Daxin Jiang, and Nan Duan. 2022. NÜWA: Visual Synthesis Pre-training for Neural Visual World Creation. In *Proceedings of the European Conference on Computer Vision (ECCV)*.
[51] Jay Zhangjie Wu, Yixiao Ge, Xintao Wang, Stan Weixian Lei, Yuchao Gu, Yufei Shi, Wynne Hsu, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. 2023. Tune-A-Video: One-Shot Tuning of Image Diffusion Models for Text-to-Video Generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
[52] Jinbo Xing, Menghan Xia, Yong Zhang, Haoxin Chen, Xintao Wang, Tien-Tsin Liu, and Ying Shan. 2024. DynamiCrafter: Animating Open-Domain Images with Video Diffusion Priors. *Proceedings of the European Conference on Computer Vision (ECCV)* (2024).
[53] Shuai Yang, Zhifei Chen, Yijun Wu, Jianming Shi, Zhe Xu, Tong Chen, and Fan Bai. 2024. SeedStory: Multimodal Long Story Generation with Large Language and Diffusion Models. *arXiv preprint arXiv:2407.08683* (2024).
[54] Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. 2024. CogVideoX: Text-to-Video Diffusion Models with an Expert Transformer. *arXiv preprint arXiv:2408.06072* (2024).
[55] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. *arXiv preprint arXiv:2308.06721* (2023).
[56] Xiang Zheng et al. 2025. LoCoT2V-Bench: A Benchmark for Long-Form and Complex Text-to-Video Generation. *arXiv preprint arXiv:2510.26412* (2025).
[57] Zangwei Zheng, Xiangyu Peng, Tianji Yang, Chenhui Cheng, Shenggui Wang, Ang Li, Bin Fang, Yang Li, and Yang You. 2024. Open-Sora: Democratizing Efficient Video Production for All. *arXiv preprint arXiv:2412.00131* (2024).
[58] Yupeng Zhou, Daquan Zhou, Ming-Ming Cheng, Jiashi Feng, and Qibin Hou. 2024. StoryDiffusion: Consistent Self-Attention for Long-Range Image and Video Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*.